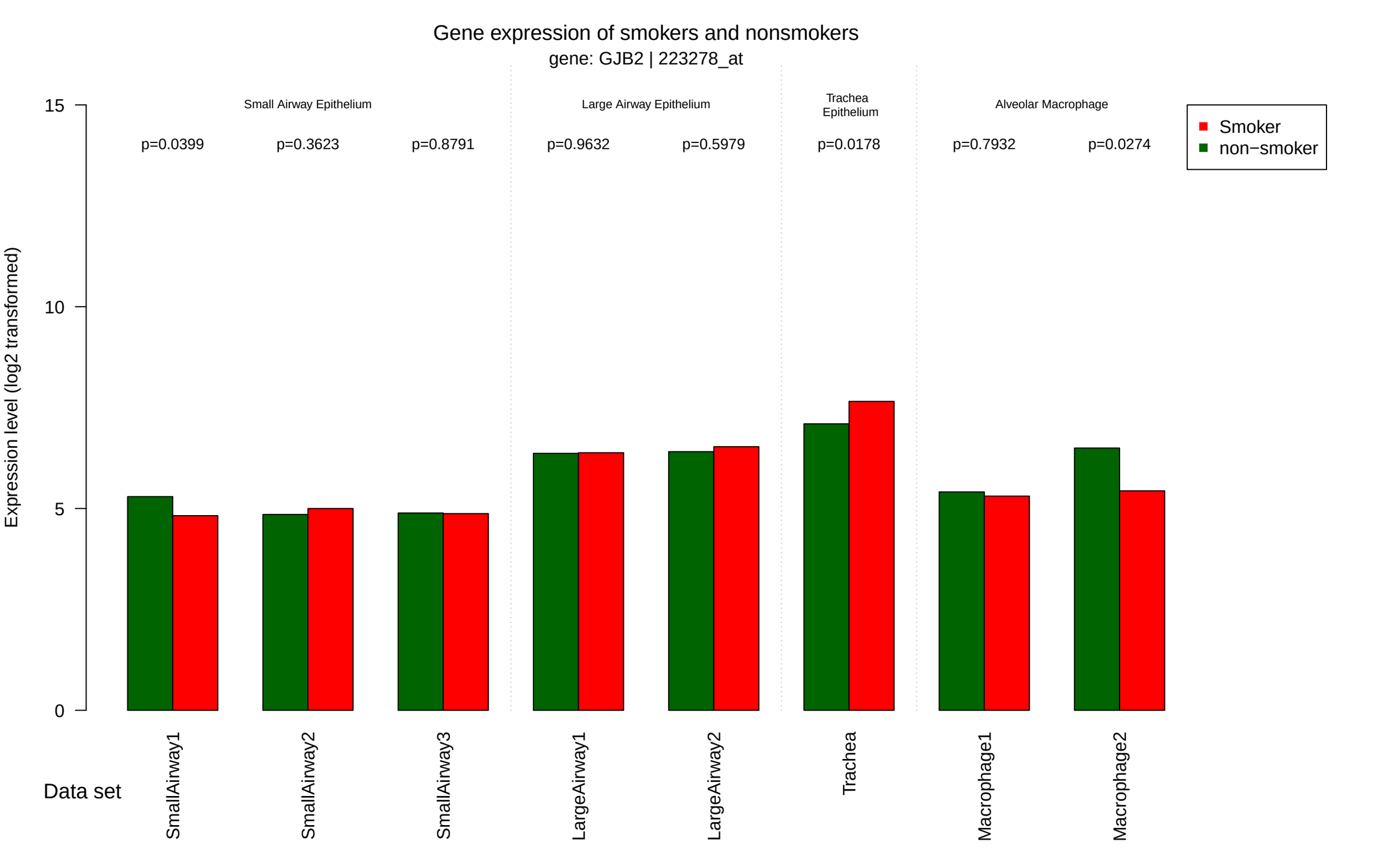
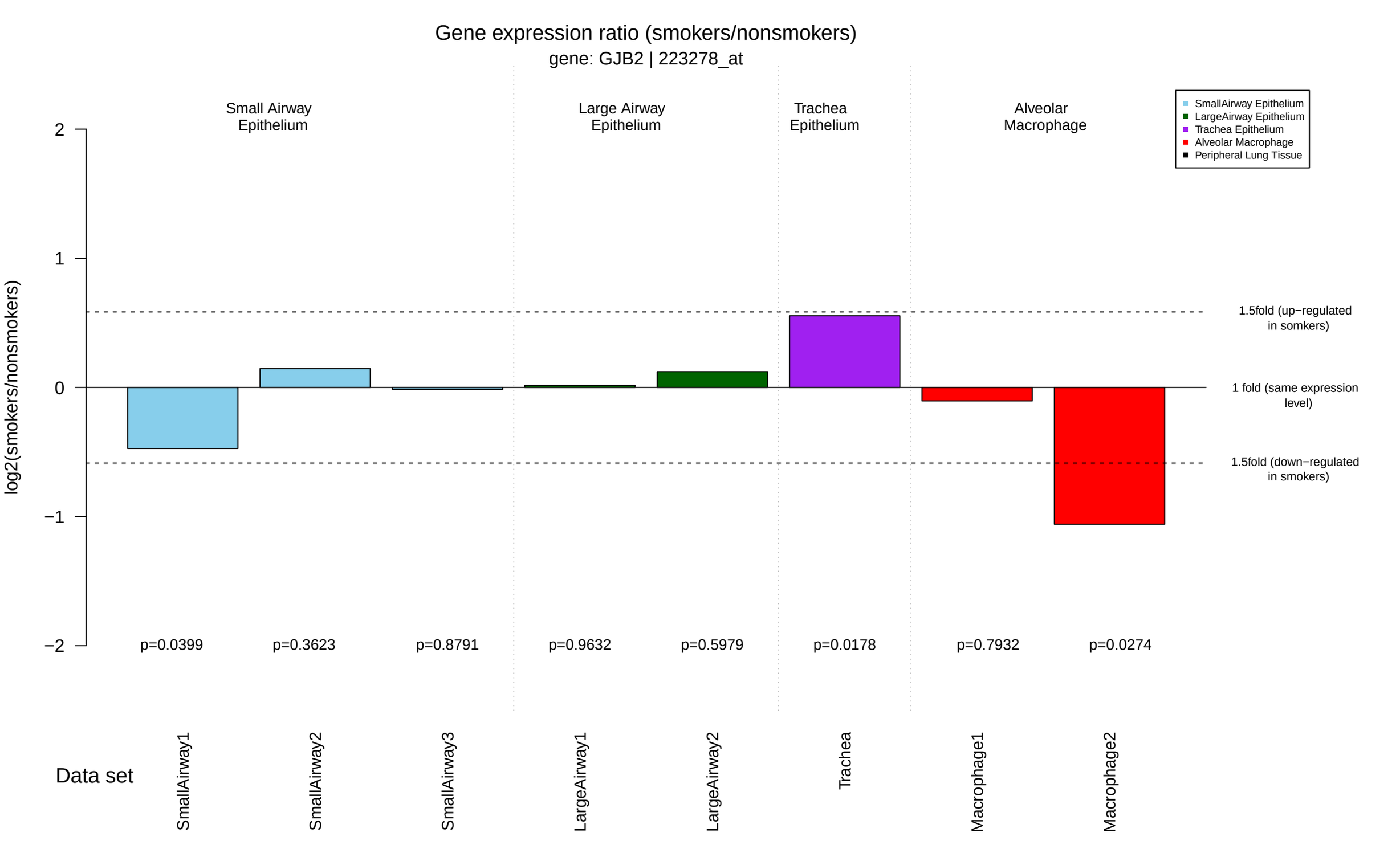
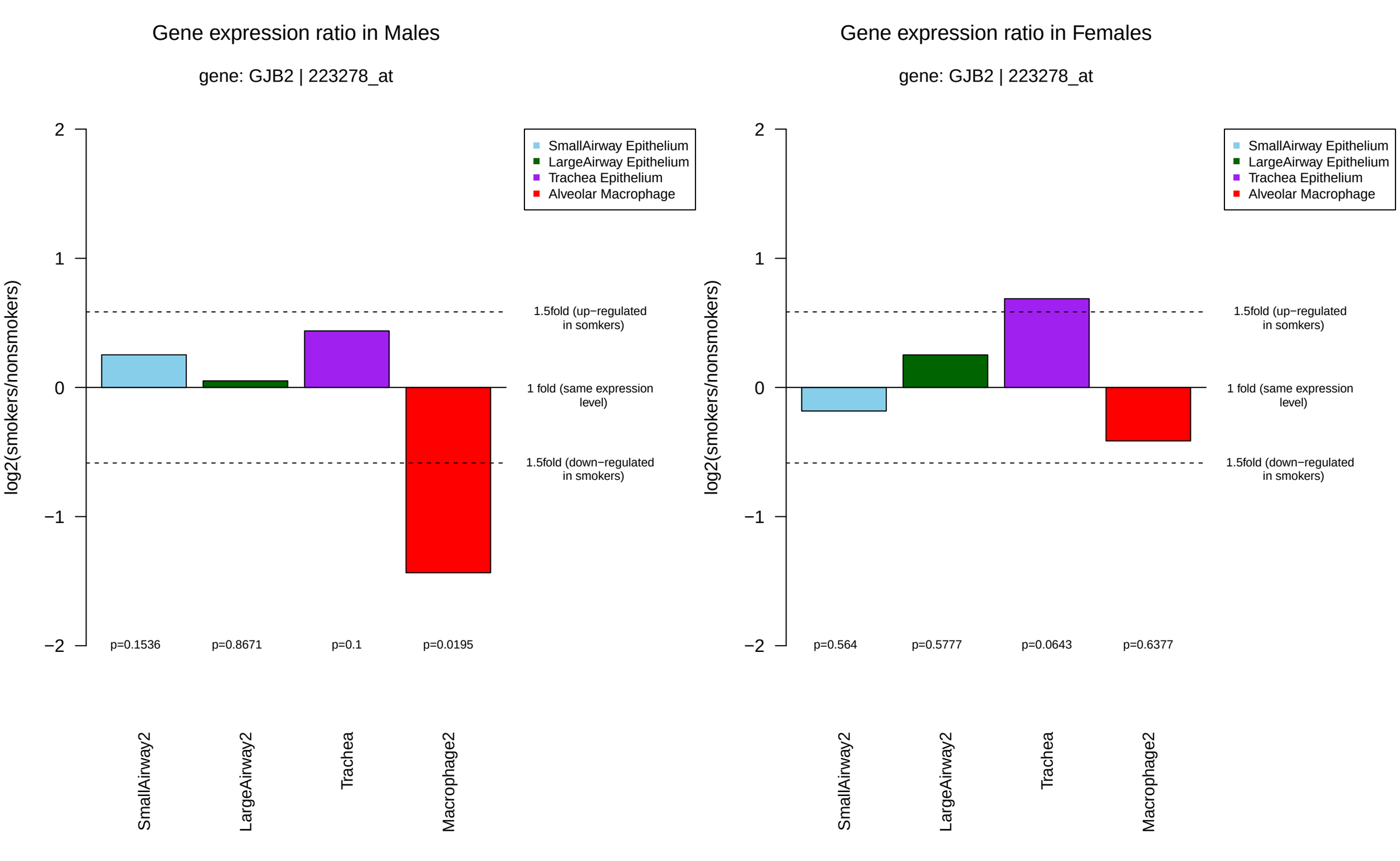
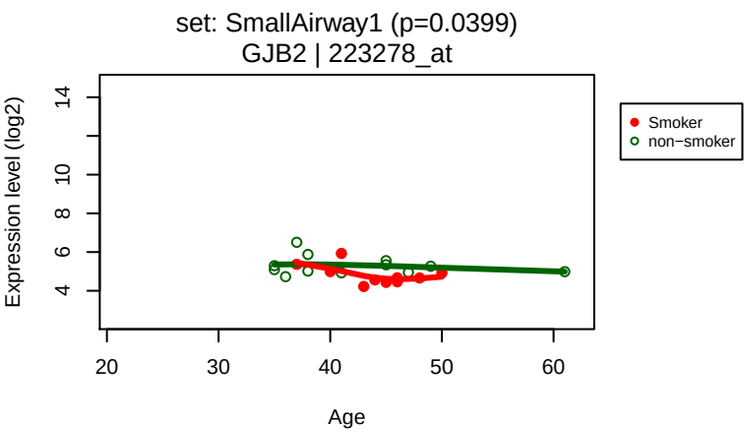
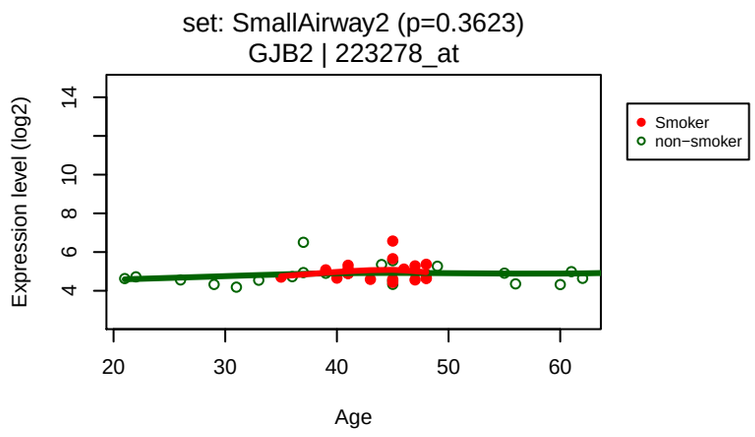
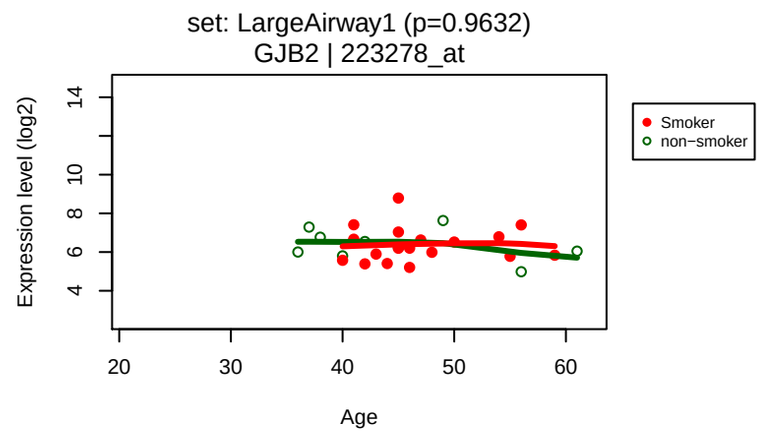

Gene expression of smokers and nonsmokers
gene: GJB2 | 223278_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0399
p=0.3623
p=0.8791
p=0.9632
p=0.5979
p=0.0178
p=0.7932
p=0.0274
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

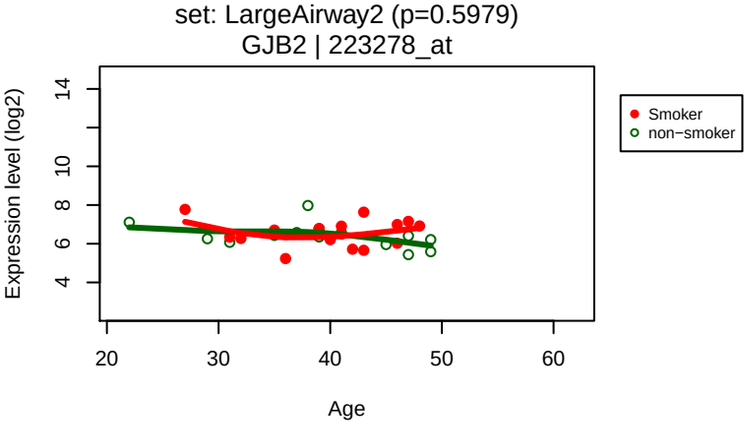

Gene expression ratio (smokers/nonsmokers)
gene: GJB2 | 223278_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0399
p=0.3623
p=0.8791
p=0.9632
p=0.5979
p=0.0178
p=0.7932
p=0.0274
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

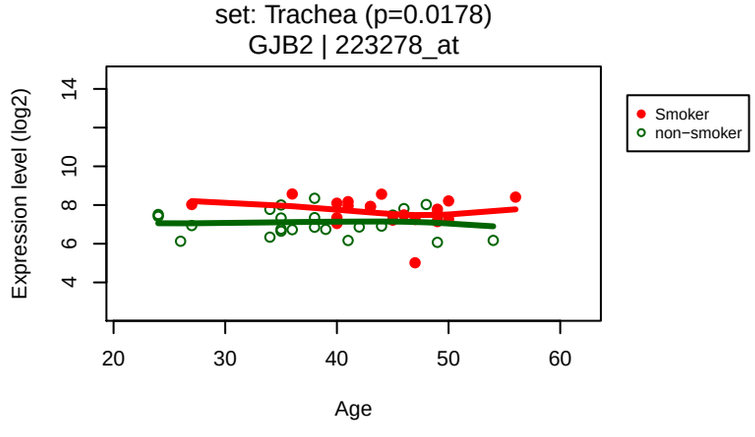

Gene expression ratio in Males
gene: GJB2 | 223278_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1536
p=0.8671
p=0.1
p=0.0195
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GJB2 | 223278_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.564
p=0.5777
p=0.0643
p=0.6377
SmallAirway2
LargeAirway2
Trachea
Macrophage2

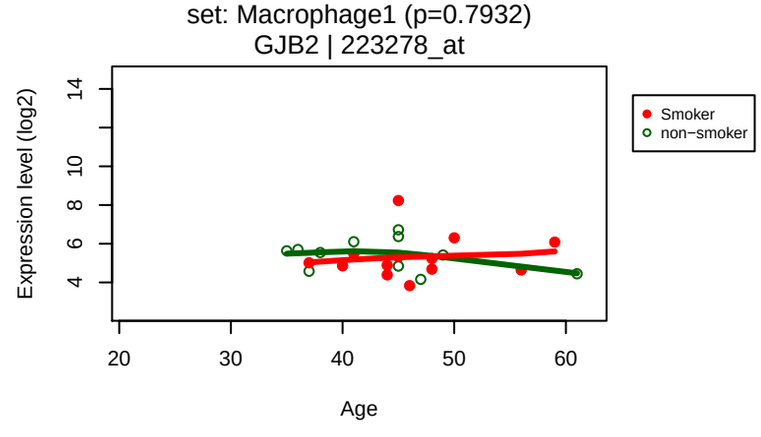

set: SmallAirway1 (p=0.0399)
GJB2 | 223278_at
Expression level (log2)
Age
Smoker
non−smoker

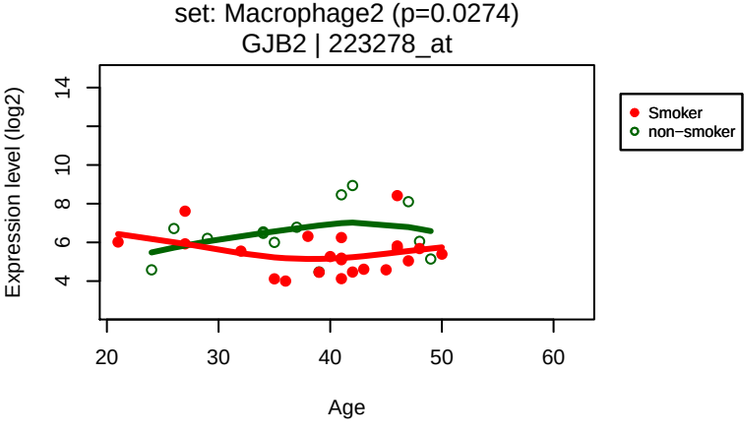

set: SmallAirway2 (p=0.3623)
GJB2 | 223278_at
Expression level (log2)
Age
Smoker
non−smoker

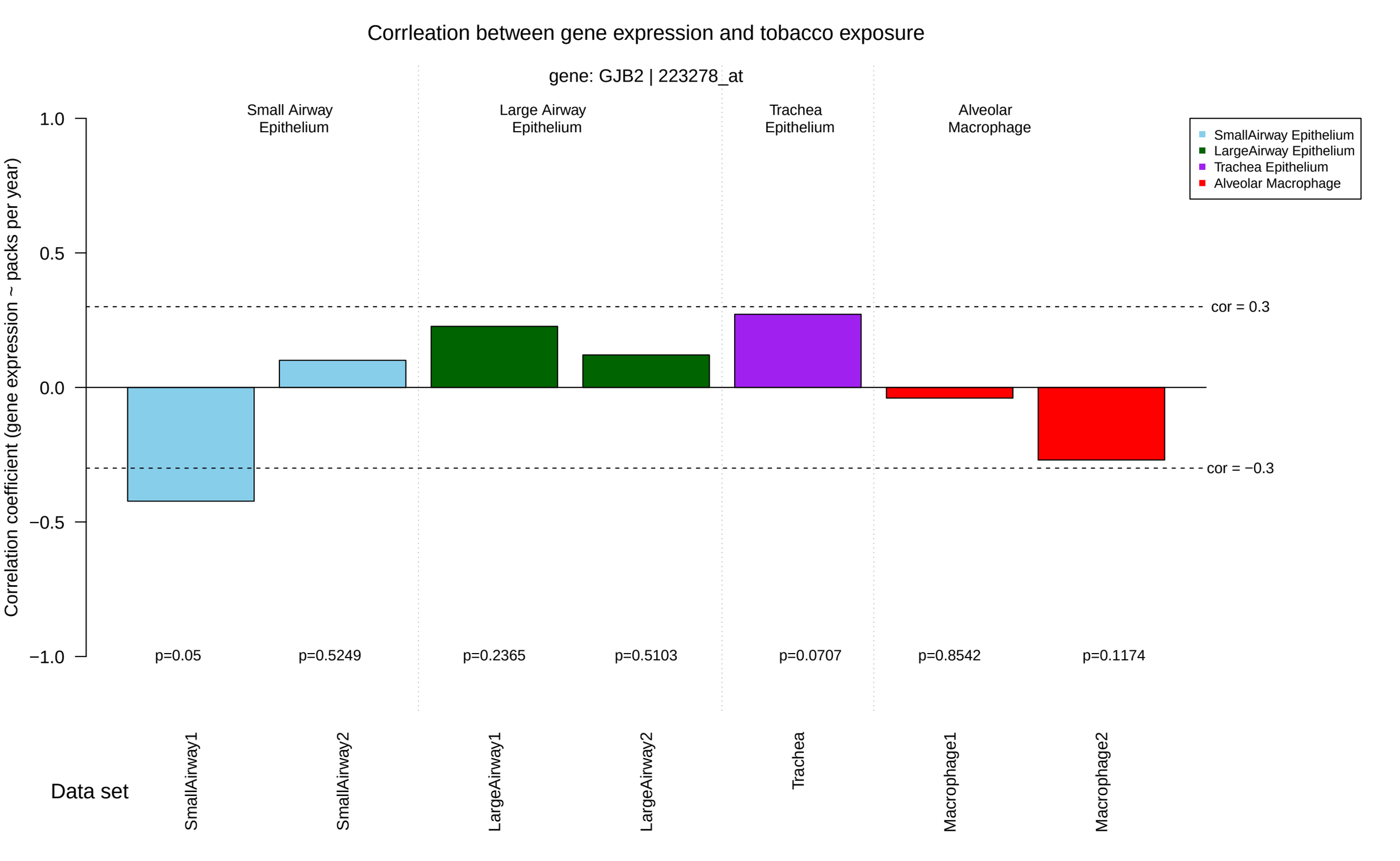

set: LargeAirway1 (p=0.9632)
GJB2 | 223278_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5979)
GJB2 | 223278_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0178)
GJB2 | 223278_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7932)
GJB2 | 223278_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0274)
GJB2 | 223278_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GJB2 | 223278_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.05
p=0.5249
p=0.2365
p=0.5103
p=0.0707
p=0.8542
p=0.1174
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2